

# OSSP PRACTICAL EXPERIMENTS

Roll Number: 2520030187 Harshit Sajjanapu

Program-1:

Code:

```
GNU nano 8.7.1
#include <stdio.h>
int main(){
printf("Welcome to OS \n");
return 0;
}
```

Output:

```
hershy@Ubuntu:~/C programs$ nano welcome.c
hershy@Ubuntu:~/C programs$ gcc welcome.c -o welcome
hershy@Ubuntu:~/C programs$ ./welcome
Welcome to OS
hershy@Ubuntu:~/C programs$
```

Program-2:

Code:

```
GNU nano 8.7.1
#include <stdio.h>
int main(){
char name[30];
printf("Enter your name\n");
scanf("%s",name);
printf("Welcome %s\n",name);
return 0;
}
```

Output:

```
hershy@Ubuntu:~/C programs$ nano name.c
hershy@Ubuntu:~/C programs$ gcc name.c -o name
hershy@Ubuntu:~/C programs$ ./name
Enter your name
Harshit
Welcome Harshit
hershy@Ubuntu:~/C programs$
```

Program-3:

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
int main(){
printf("Process ID =%d\n",getpid());
return 0;
}
```

Output:

```
hershy@Ubuntu:~/C programs$ ./getID
Process ID =22601
hershy@Ubuntu:~/C programs$
```

Program-4:

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
int main(){
fork();
printf("Hello\n");
return 0;
}
```

Output:

```
hershy@Ubuntu:~/C programs$ nano fork.c
hershy@Ubuntu:~/C programs$ gcc fork.c -o fork
hershy@Ubuntu:~/C programs$ ./fork
Hello
Hello
hershy@Ubuntu:~/C programs$
```

#### Program-5:

##### Code:

```
#include <stdio.h>
#include <unistd.h>
int main(){
    pid_t pid;
    pid =fork();
    if(pid==0){
        printf("I am  child process\n");
    }
    else{
        printf("I am parent process\n");
    }
    return 0;
}
```

##### Output:

```
bash: ./cp_prog: is a directory
hershy@Ubuntu:~/C programs$ ./cp_prog
I am parent process
I am  child process
hershy@Ubuntu:~/C programs$
```



### Program-6:

#### Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
int main(){
    pid_t pid;
    pid=fork();
    if(pid==0){
        printf("Child  pid=%d\n",getpid());
        printf("Parent pid=%d\n",getpid());
    }
    else{
        printf("Parent pid=%d\n",getpid());
    }
    return 0;
}
```

#### Output:

```
hershy@Ubuntu:~/C programs$ nano childParentPID.c
hershy@Ubuntu:~/C programs$ gcc childParentPID.c -o childParentPID
hershy@Ubuntu:~/C programs$ ./childParentPID
bash: ./childParentPID: No such file or directory
hershy@Ubuntu:~/C programs$ ./childParentPID
Parent pid=6318
Child  pid=6319
Parent pid=6319
hershy@Ubuntu:~/C programs$
```

### Program-7:

#### Code:

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h> //we can use wait/sys for this

int main() {
    pid_t pid;
    pid = fork();
    if (pid == 0) {
        printf("Child Running\n");
    }
    else {
        wait(NULL);
        printf("Parent running After Child\n");
    }
    return 0;
}
```

Output:

```
hershy@Ubuntu:~/C programs$ nano prac2.c
hershy@Ubuntu:~/C programs$ ./wait
Child Running
Parent running After Child
hershy@Ubuntu:~/C programs$
```